

Sailesh Dwivedy

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(Open to Relocate)

SUMMARY

Machine Learning Engineer with 5+ years of experience designing and deploying scalable **ML**, **NLP**, and **GenAI** solutions for Fortune 500 clients across **healthcare**, **finance**, and **research** domains. Expertise in developing end-to-end **ML** pipelines, **fine-tuning** LLMs, **RAG**, and building **agentic AI** workflows. **Actively seeking opportunities** to lead complex data science projects and deliver high-impact AI solutions.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, PyTorch, Scikit-learn, Pandas, NumPy, Spark
- **Agentic AI and RAG:** LlamaIndex, LangChain, FAISS, Vector Stores
- **NLP and GenAI:** Hugging Face, Transformers, Fine-tuning, PEFT (LoRA/QLoRA), RLHF, Prompt Engineering
- **Machine Learning and Deep Learning:** Supervised ML, Classification (Imbalanced), Regression, Ensemble Models, Neural Networks
- **MLOps & Deployment:** Docker, FastAPI, AWS (S3, SageMaker), GitHub, MLflow, Weights & Biases
- **Data Analysis and Visualization:** Exploratory Data Analysis (EDA), Tableau, Plotly, Matplotlib

PUBLICATIONS, CERTIFICATIONS, AWARDS

- **Logit-Based Losses Limit the Effectiveness of Feature Knowledge Distillation**, NeurIPS Workshop 2025
- AI Agents Fundamentals, Hugging Face, [[Certificate](#)]
- Generative AI with Large Language Models, Deep Learning.AI, and Amazon Web Services, [[Certificate](#)]
- Event-Driven Agentic Document Workflows, DeepLearning.AI and LlamaIndex, [[Certificate](#)]
- Awarded as Best Performer of the team for 5 consecutive quarters, *Associated with Highradius Technology*
- Winner at Smart India Hackathon 2019, *Associated with the Government of India*

PROFESSIONAL EXPERIENCE

NEUROSPRING

Machine Learning Engineer

Remote

Jun 2025 - Present

Delivering ML and GenAI solutions in the healthcare domain, focusing on speech analysis and clinical text generation. Building end-to-end AI pipelines, integrating interpretability and retrieval systems to assist medical professionals.

- Achieved **80% F1** in slurred speech detection by building **ML pipelines** and **85% F1** using **deep learning** with **PyTorch**, tracked with **MLflow**, deployed via **FastAPI/Docker** on **GCP**.
- Integrated **explainability frameworks** through **SHAP** to provide transparent, interpretable AI outputs for clinical decision support, facilitating **stakeholder trust** and **regulatory compliance**.
- **Fine-tuned BioMistral-7B** using **LoRA** and **4-bit quantization**, in PyTorch/Hugging Face, enabling paramedics to generate clinical notes, tracked **50+ runs** with **Weights & Biases**.
- Developing a **Retrieval-Augmented Generation** pipeline to help paramedics access clinical guidelines.
- Building RAG with **ingestion**, **chunking**, **semantic vector search**, **reranking**, and citation-based answers, and validated against expert-reviewed golden test questions.

IMAGE AND VIDEO COMPUTING LAB, UNIVERSITY OF COLORADO, BOULDER

Graduate AI/ML Researcher

Colorado, US

May 2024 - Dec 2024

Improved CNN and ViT performance in image classification benchmarks by developing novel teacher-student optimization methods for knowledge distillation, resulting in faster convergence and higher accuracy.

- Co-authored **NeurIPS Workshop 2025** paper **Logit-Based Losses Limit the Effectiveness of Feature Knowledge Distillation**, improving CNN and ViT accuracy by **15%** with a novel logit-free KD framework.
- Achieved **80% faster convergence** on CIFAR/Tiny ImageNet by designing a novel **Knowledge Quality (KQ)** metric to automatically select optimal teacher layers, implemented in PyTorch.

- Conducted **extensive ablation** experiments (**50+ runs**), showing that all three KQ components were necessary for optimal student convergence.

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

NLP Research Associate

Odisha, India
Feb 2023 - Aug 2023

Conducted applied research on transformer-based document parsing to improve the accuracy of information extraction from business documents. The work demonstrated measurable accuracy gains over existing baselines.

- Evaluated **LayoutLMv3**, a **Transformer-based** document parser, against LayoutLMv2 and OCR-based baselines on document datasets, showing **5 to 10%** accuracy improvements in field extraction tasks.
- Tracked and compared **50+ experiments** with **MLflow**, monitoring model comparison, loss convergence, and parameter tuning to ensure reproducibility and reliability during training.

HIGHRADIUS TECHNOLOGIES PRIVATE LIMITED

AI/ML Engineer - Team Lead (FinTech)

Odisha, India | Texas, USA
Jul 2019 - Feb 2023

Client: Red Bull, Danone, Hershey's, Staples, Pavion

Led a 10-member team delivering end-to-end production-grade AI/ML solutions for 15+ Fortune 500 clients, improving revenue recovery and operational efficiency for fraud detection, payment prediction, cash forecasting, and automating model lifecycle workflows, deployed on AWS.

Client: Redbull, Danone

Project 1: Deduction validity prediction - Designed and deployed ML predictors for deduction fraud detection, improving revenue recovery and analyst efficiency, and accelerating dispute resolution with explainable predictions.

- **Increased dollar recovery** by **12%** and **reduced** manual **research effort** by **25%**, building **classification** models (LightGBM, Random Forest) with **85% recall** for **fraud detection**, ensuring **SHAP** interpretability.
- **Automated** low-dollar deduction **write-offs** and **prioritized research worklists** through AI action suggestions, accelerating dispute resolution and improving operational efficiency.
- Collaborated with **client finance teams**, driving presentations to **align model outputs with business workflows**, ensuring adoption and trust in AI-driven fraud detection.

Client: Hershey's, Staples

Project 2: Payment date prediction - Developed an AI framework analyzing 20+ customer risk factors to predict invoice payment dates and prioritize delinquent accounts with proactive collections strategies, improving cash flow.

- **Increased on-time collections** by **17%** by building an **AI prioritization system** using payment history features to generate **personalized worklists**, enabling targeted follow-ups with high-risk customers.
- Built the prioritization system by **predicting** invoice **payment dates** with delay buckets, using **regression models** (XGBoost, Random Forest) validated with **RMSE**, improving collector efficiency by **20%**.

Client: Pavion

Project 3: Predictive Cash Forecasting - Improved treasury decision-making by building an AI-driven forecasting solution that provides real-time liquidity visibility, enabling smarter investments, timely debt repayments, and proactive risk management.

- Achieved **90%+ accuracy** in **cash flow forecasting** by building **time-series** pipelines (**SARIMA, Prophet**) on payables and receivables data, enabling treasury teams to **plan investments** more effectively.
- Improved **liquidity planning** by **automating forecasting workflows** and validating models via backtesting.

Project 4: Machine Learning Pipeline Automation - Improved client delivery efficiency by automating end-to-end ML implementation and deployment pipelines across various products.

- Improved **model implementation** and **deployment** timelines by **40%** by automating preprocessing, feature engineering, training, and evaluation pipelines in **Amazon SageMaker**, with **MLflow** tracking.
- **Led 10 member team** across Order-to-Cash products and supported stakeholder decisions with **Tableau** insights.
- Collaborated with **cross-functional teams** to translate business requirements into scalable AI solutions, ensuring strategic alignment across technical, product, and operations stakeholders.

PROJECTS

ChatPDF-RAG: Chat with Your PDFs using LangChain and AWS Bedrock [[Video Demo](#) | [Source Code](#)]

Tech Stack: Python, LangChain, FAISS, AWS Bedrock (Titan Embeddings, Claude 3 Haiku), pypdf

- Developed a **Retrieval-Augmented Generation** system (LangChain, FAISS, Bedrock) enabling concise, grounded PDF question answering.
- Built **chunking, retry logic, deduplication**, and **source attribution**, ensuring production-grade readiness.

ImageSense AI: Multimodal Image Retrieval System [[Video Demo](#) | [Source Code](#)]

Tech Stack: Python, CLIP, FAISS, LLaMA 3.2, Transformers, PyTorch

- Developed an AI-powered **multimodal text-to-image retrieval pipeline** that enhances search relevance using CLIP-based image embeddings, FAISS vector indexing and LLMs for query expansion.
- Integrated LLaMA 3.2 via Ollama for semantic grounding and query expansion, improving search relevance.

EDUCATION

UNIVERSITY OF COLORADO BOULDER

Master’s in Computer Science (GPA: 3.90/4)

Colorado, US

Aug 2023 – May 2025

KIIT UNIVERSITY

Bachelor’s in Computer Science

Odisha, India

May 2016 – May 2020